KAIZEN MATHS

Lesson starter

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Root Location and Iteration

Name

S1

Evaluate the function at both ends of the interval and use the sign change to locate a root.

Show that $x^2 - 11 = 0$ has a root in the interval [3, 4].

S2

Use the midpoint to choose a smaller interval containing the root.

For $f(x) = x^2 - 17$, one root lies in [4, 5]. Use one bisection step to find the next interval.

S3

Apply the iteration formula carefully and give each approximation to the stated accuracy.

Use $x_{n+1} = \frac{1}{2}(x_n + \frac{3}{x_n})$ with $x_0 = 1$ to find x_1 and x_2.

S4

Evaluate the function at both ends of the interval and use the sign change to locate a root.

Show that $x^2 - 26 = 0$ has a root in the interval [5, 6].

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Show that $x^2 - 26 = 0$ has a root in the interval [5, 6].

ALG-D1B0-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Root Location and Iteration

Name _____

EA1

Use the midpoint to choose a smaller interval containing the root.

For $f(x) = x^2 - 22$, one root lies in $[4, 5]$. Use one bisection step to find the next interval.

EA2

Apply the iteration formula carefully and give each approximation to the stated accuracy.

Use $x_{n+1} = \frac{1}{2}(x_n + \frac{6}{x_n})$ with $x_0 = 2$ to find x_1 and x_2.

KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Root Location and Iteration

Name _____

EB1

Evaluate the function at both ends of the interval and use the sign change to locate a root.

Show that $x^2 - 22 = 0$ has a root in the interval $[4, 5]$.

KAIZEN MATHS

Exit ticket

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Root Location and Iteration

Name _____

EA1

Use the midpoint to choose a smaller interval containing the root.

For $f(x) = x^2 - 22$, one root lies in $[4, 5]$. Use one bisection step to find the next interval.

EA2

Apply the iteration formula carefully and give each approximation to the stated accuracy.

Use $x_{n+1} = \frac{1}{2}(x_n + \frac{6}{x_n})$ with $x_0 = 2$ to find x_1 and x_2.

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Show your working

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Exit ticket

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Root Location and Iteration

Name _____

EB1

Evaluate the function at both ends of the interval and use the sign change to locate a root.

Show that $x^2 - 22 = 0$ has a root in the interval $[4, 5]$.

ALG-D1B0-L1 · EB

Show your working

Teacher answers & working

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule · Root Location and Iteration

ALG-D1B0-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Evaluate the function at both ends of the interval and use the sign change to locate a root.

Show that $x^2 - 11 = 0$ has a root in the interval [3, 4].

Answer:

A root lies in [3, 4].

Working:

Let $f(x) = x^2 - 11$.

$f(3) = 3^2 - 11 = -2$.

$f(4) = 4^2 - 11 = 5$.

f(3) is negative and f(4) is positive, so the function changes sign on [3, 4].

Since f is continuous, there is a root in [3, 4].

S2

Use the midpoint to choose a smaller interval containing the root.

For $f(x) = x^2 - 17$, one root lies in [4, 5]. Use one bisection step to find the next interval.

Answer:

Next interval: [4, 4.5].

Working:

The midpoint of [4, 5] is $\frac{4+5}{2} = 4.5$.

$f(4) = -1$.

$f(4.5) = (4.5)^2 - 17 = 3.25$.

$f(5) = 8$.

The sign changes between x = 4 and x = 4.5, so the root is in [4, 4.5].

S3

Apply the iteration formula carefully and give each approximation to the stated accuracy.

Use $x_{n+1} = \frac{1}{2}(x_n + \frac{3}{x_n})$ with $x_0 = 1$ to find x_1 and x_2.

Answer:

$x_1 = 2, x_2 = 1.75$.

Working:

Substitute $x_0 = 1$ into the iteration formula.

$x_1 = \frac{1}{2}(1 + \frac{3}{1}) = 2$.

Now substitute $x_1 = 2$ back into the same formula.

$x_2 = \frac{1}{2}(2 + \frac{3}{2}) = 1.75$.

The sequence is moving towards an approximation to √3.

Teacher answers & working

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S4

Evaluate the function at both ends of the interval and use the sign change to locate a root.

Show that $x^2 - 26 = 0$ has a root in the interval [5, 6].

Answer:

A root lies in [5, 6].

Working:

Let $f(x) = x^2 - 26$.

$f(5) = 5^2 - 26 = -1$.

$f(6) = 6^2 - 26 = 10$.

f(5) is negative and f(6) is positive, so the function changes sign on [5, 6].

Since f is continuous, there is a root in [5, 6].

EA1

Use the midpoint to choose a smaller interval containing the root.

For $f(x) = x^2 - 22$, one root lies in [4, 5]. Use one bisection step to find the next interval.

Answer:

Next interval: [4.5, 5].

Working:

The midpoint of [4, 5] is $\frac{4+5}{2}$ = 4.5.

$f(4) = -6$.

$f(4.5) = (4.5)^2 - 22 = -1.75$.

$f(5) = 3$.

The sign changes between x = 4.5 and x = 5, so the root is in [4.5, 5].

EA2

Apply the iteration formula carefully and give each approximation to the stated accuracy.

Use $x_{n+1} = \frac{1}{2}(x_n + \frac{6}{x_n})$ with $x_0 = 2$ to find x_1 and x_2.

Answer:

$x_1 = 2.5, x_2 = 2.45$.

Working:

Substitute $x_0 = 2$ into the iteration formula.

$x_1 = \frac{1}{2}(2 + \frac{6}{2}) = 2.5$.

Now substitute $x_1 = 2.5$ back into the same formula.

$x_2 = \frac{1}{2}(2.5 + \frac{6}{2.5}) = 2.45$.

The sequence is moving towards an approximation to √6.

Teacher answers & working

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EB1

Evaluate the function at both ends of the interval and use the sign change to locate a root.

Show that $x^2 - 22 = 0$ has a root in the interval [4, 5].

Answer:

A root lies in [4, 5].

Working:

Let $f(x) = x^2 - 22$.

$f(4) = 4^2 - 22 = -6$.

$f(5) = 5^2 - 22 = 3$.

f(4) is negative and f(5) is positive, so the function changes sign on [4, 5].

Since f is continuous, there is a root in [4, 5].